

	Exam	ent	Evaluation	Examinations		
CO 1	✓			✓		
CO 2	✓	✓		✓		
CO 3	✓			✓		
CO 4		✓		✓		
CO 5		✓		✓		
CO 6	✓					
Programme		B. Sc. Statistics				
Course Code		STA6FS113				
Course Title		Basic research methodology				
Type of Course		SEC				
Semester		VI				
Academic Level		100-199				
Course Details		Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
		3	3	-	-	45
Pre-requisites		HSE level Mathematics Course				
Course Summary		To make students aware of research methodology.				

Course outcome CO

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Identify gaps, contradictions and areas for further exploration within the existing body of literature.	R	F	Instructor-created exams / Quiz
CO2	Ethically and accurately cite article references in accordance with academic integrity standards.	U	C	Instructor-created exams / Home Assignments
CO3	Develop a coherent and logically structured thesis report that adheres to academic conventions and standards.	R	P	Seminar Presentation / Group Tutorial Work

CO4	Construct text, equations, figures, tables and references in accordance with academic standards and publication guidelines and analyze data to help entrepreneurial decisions using critical thinking skills.	U	C	Instructor-created exams / Home Assignments /Seminar Presentation
CO5	Ethically and responsibly apply numerical methods in research, acknowledging limitations and uncertainties.	Ap	F	Seminar Presentation /Quiz
CO6	Identify appropriate case study designs and data collection methods for specific research questions and contexts.	R	P	Seminar Presentation/ Group Tutorial Work
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (48 +12)	Marks (70)
I		Research in Statistics	12	15
	1	Concept of Research in Statistics-Importance and Need for Research Ethics	2	
	2	Selection of Topic for Research-Research schedules	3	
	3	Review of Literature and its Use in Designing a Research Work-	3	
	4	Mode of Literature Survey	2	
	5	Thesis Writing – Computer Application in Scientific Research	2	
		Sections from References:		
II			12	15
	6	Scientific Word Processing with LaTeX	2	
	7	Article, References	2	
	8	Thesis Report and Slide Preparation	2	

	9	Statistical Programming with R: Arrays and Matrices-Lists	2	
	10	Data Frames-Grouping, Loops and Conditions	2	
	11	Probability Distributions and Statistical Models in R.	2	
	Sections from References:			
III			15	25
	112	Simulation: Concepts and Advantages of Simulation-	2	
	13	Event Type Simulation- Random Variable Generation-U(0,1)	2	
	14	Exponential, Gamma and Normal Random Variables – Monte Carlo Integration.	3	
	15	The MCMC Principle,	3	
	16	Algorithms and its Variants	2	
	17	Bootstrap Methods.	3	
	Sections from References:			
IV			9	15
	18	Computer Oriented Numerical Methods	2	
	19	Algorithms for Solving Algebraic Equations	2	
	20	Algorithms for Solving Transcendental Equations	1	
	21	Numerical Integration	2	
	22	Matrix operations.	2	
	Sections from References:			
V	Open Ended Module:		12	
	1	Analysis of a case study	12	
	Sections from References:			

Books and References:

7. Anderson, J., Durston, B.H., Poole, M. (1970) Thesis and Assignment Writing. Wiley Eastern. Ltd., New Delhi.
8. Beveridge, B. (1979) The Art of Scientific Investigation. W.E. Norton & Co., New York. Braun, J., Duncan, W. and Murdock, J. (2008) A First Course in Statistical Programming with R. Cambridge University Press, London.
9. Chambers, J. (2008) Software for Data Analysis: Programming with R. Springer, New York.
10. Crewley, M.J. (2007) The R-Book. John Wiley, New York.
11. Dalgaard, P.(2008) Introductory Statistics with R. Springer Science, New York.
12. Ghosh, J.K., Mitra, S.K. and Parthasarathy, K. R.(1992) Glimpes of India's Statistical Heritage. Wiley Eastern Limited, New Delhi.
13. Hald, A.(1998) A History of Mathematical Statistics from 1750 to 1930. John Wiley & Sons, New York.
14. Kantiswarup, S., Gupta P.K. and Man Mohan (2008) Operations Research. Sultan Chand & Sons, New Delhi.
15. Kothari, C. (2005) Research Methodology. New Age International. Publishers, New York.
16. Lamport, L. (1999) LATEX: A Document Preparation System. Addison, nd Wesley, 2
17. Pannerselvan,R. (2006) Research Methodology. Prentice-Hall of India. Pvt., NewDelhi.
18. Robert, C.P. and Casella, G. (2004) Monte Carlo Statistical Methods. Springer Science, New York.
19. Venkataraman, M.K. (1998) Numerical Methods in Science and Engineering. The National Publishing Company, Chennai.

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	3	-	-	-	2	2	3	-	-	-	-
CO 2	-	-	2	-	-	-	-	-	-	3	-	2
CO 3	-	2	-	3	2	3	1	2	3	-	-	-
CO 4	3	2	2	-	-	3	3	1	-	3	3	-
CO 5	2	-	2	-	-	-	1	-	-	3	-	3
CO 6	-	-	3	-	-	2	1	2	-	3	-	-

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

4. Quiz / Assignment/ Quiz/ Discussion / Seminar
5. Midterm Exam
6. Programming Assignments (20%)
7. Final Exam (70%)

Mapping of COs to Assessment Rubrics:

	Internal Exam	Assignm ent	Project Evaluation	End Semester Examinations
CO 1	✓			✓
CO 2	✓			✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6			✓	

